

Ncat / Netcat — Practical Tool Guide

Connect, listen, grab banners, use TLS, test connectivity, and transfer files inside an authorized lab.

➤ [Abdullah Nasser Almsan — Personal Pentest Reference](#)

Usage note: This guide focuses on basic network plumbing and intentionally avoids shell payloads so the reference stays simple and clear.

CONNECT

Connect

BANNER

Read the service

LISTEN

Receive

VERIFY

Confirm transfer

Contents

1. Connect and Listen
2. Banner grabbing
3. Port and connectivity testing
4. File transfer inside a lab
5. When to use Ncat instead of another tool

1. Connect and Listen

Ncat/Netcat is a general tool for reading and writing TCP/UDP data.

KALI TERMINAL — TCP connection

```
ncat [TARGET] [PORT]
```

KALI TERMINAL — Listen on a port

```
ncat -l 9001
```

KALI TERMINAL — Verbose

```
ncat -v [TARGET] [PORT]
```

Kali may also provide nc. Options vary slightly between Ncat and OpenBSD netcat; use ncat --help when unsure.

2. Banner grabbing

Useful before/after Nmap to confirm that a service responds as expected.

KALI TERMINAL — Open raw TCP

```
ncat [TARGET] 21
```

Observe the banner for a service such as FTP.

KALI TERMINAL — Manual HTTP HEAD

```
printf "HEAD / HTTP/1.0\r\nHost: [HOST]\r\n\r\n" | ncat [TARGET] 80
```

KALI TERMINAL — TLS with Ncat

```
ncat --ssl [TARGET] 443
```

3. Port and connectivity testing

KALI TERMINAL — Verbose connect test

```
ncat -vz [TARGET] 22
```

KALI TERMINAL — UDP

```
ncat -u [TARGET] [PORT]
```

UDP does not provide the same simple confirmation as TCP; no response does not necessarily mean a closed port.

4. File transfer inside a lab

A direct way to understand one-way data flow.

RECEIVER

```
ncat -l 9001 > received.bin
```

SENDER

```
ncat [RECEIVER_IP] 9001 < file.bin
```

Verify the file hash after transfer when integrity matters. Do not leave the listener open when finished.

5. When to use Ncat instead of another tool

Task	Best fit	Why
Banner / single port	Ncat	Fast raw connection.
HTTP with headers/cookies	cURL	Far easier than constructing HTTP manually.
Scan many ports	Nmap	Purpose-built for enumeration.
Intercept HTTP	Burp	Better for analysis and modification.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://nmap.org/ncat/guide/>

<https://nmap.org/book/ncat-man.html>